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(54) **SEMICONDUCTOR LIGHT-EMITTING DEVICE AND PHOTOCOUPLER**

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***H10F 55/25*** (2025.01)

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(52) **U.S. Cl.**

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(58) **Field of Classification Search**

CPC .... H10H 20/818; H10H 20/852; H10H 20/01; H10H 20/824; H10H 20/857; H10H 20/812

See application file for complete search history.

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(57) **ABSTRACT**

A semiconductor light-emitting device includes a GaAs (gallium arsenide) substrate of a cubic crystal, a light-emitting layer and a multi-semiconductor layer. The light-emitting layer being provided on the GaAs substrate. The light-emitting layer includes InGaAs (indium gallium arsenide) represented by a compositional formula InxGa1-xAs (0<x<1). The multi-semiconductor layer being provided on a front surface of the GaAs substrate between the GaAs substrate and the light-emitting layer. The multi-semiconductor layer is tilted with respect to a (100) plane of the cubic crystal. The multi-semiconductor layer includes a first layer and a second layer. The first and second layers are alternately stacked in a direction perpendicular to the front

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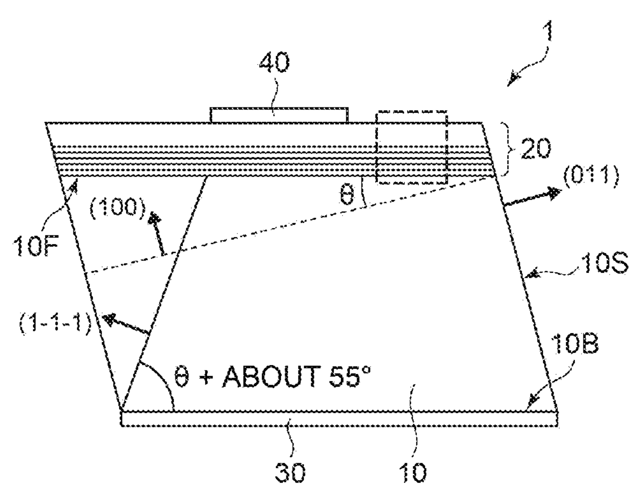

surface of the GaAs substrate. The first layer is different in composition from the second layer.

**12 Claims, 7 Drawing Sheets**

(51) **Int. Cl.**

| | |
|---|---|
| ***H10F 77/00*** | (2025.01) |
| ***H10F 77/50*** | (2025.01) |
| ***H10H 20/01*** | (2025.01) |
| ***H10H 20/812*** | (2025.01) |
| ***H10H 20/824*** | (2025.01) |
| ***H10H 20/852*** | (2025.01) |
| ***H10H 20/857*** | (2025.01) |

(52) **U.S. Cl.**

CPC .......... ***H10H 20/01*** (2025.01); ***H10H 20/812*** (2025.01); ***H10H 20/824*** (2025.01); ***H10H 20/852*** (2025.01); ***H10H 20/857*** (2025.01)

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FIG. 1A

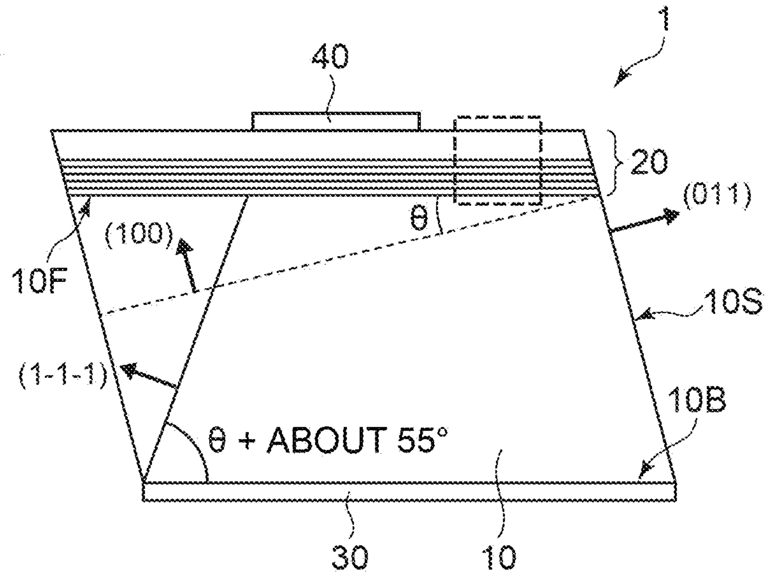
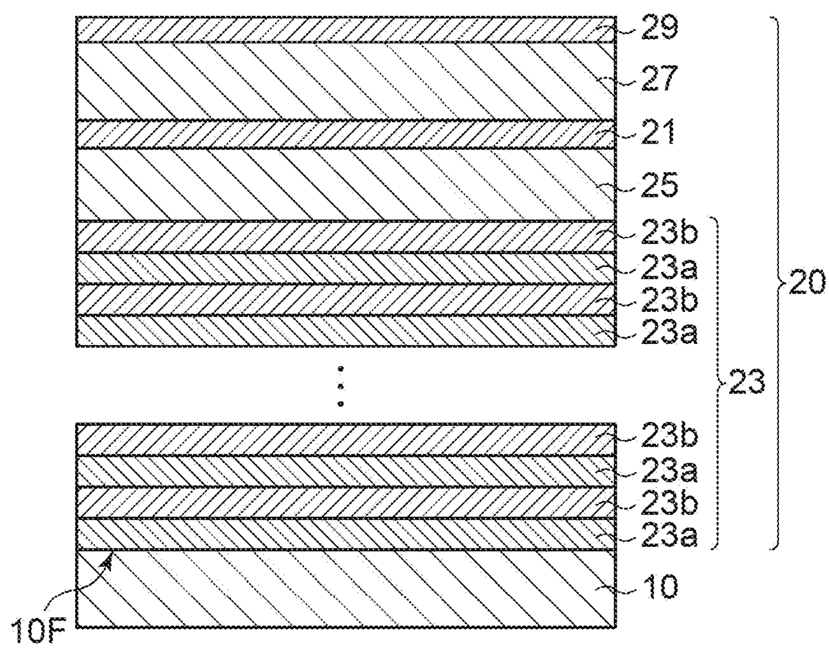

1
40
20
(011)
θ
(100)
10F
10S
(1-1-1)
10B
θ + ABOUT 55°
30
10

FIG. 1B


29
27
21
25
23b
23a
23b
23a
20
23
23b
23a
23b
23a
10
10F

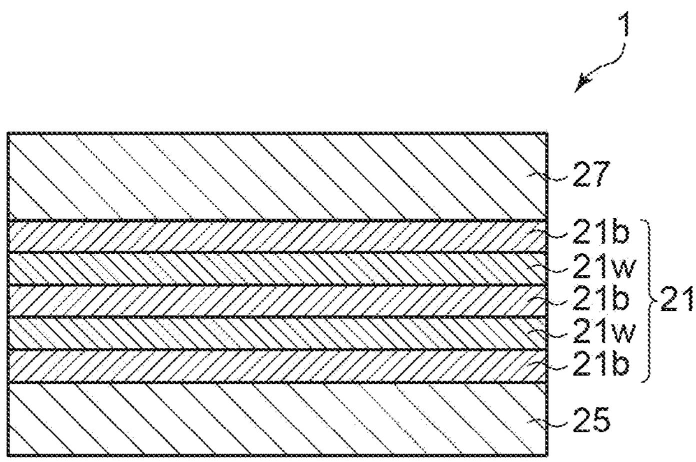



FIG. 2

FIG. 3A

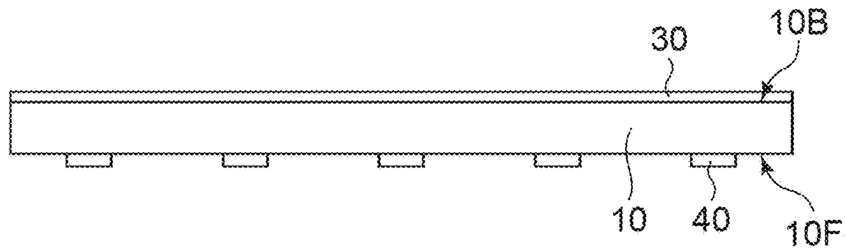



FIG. 3B

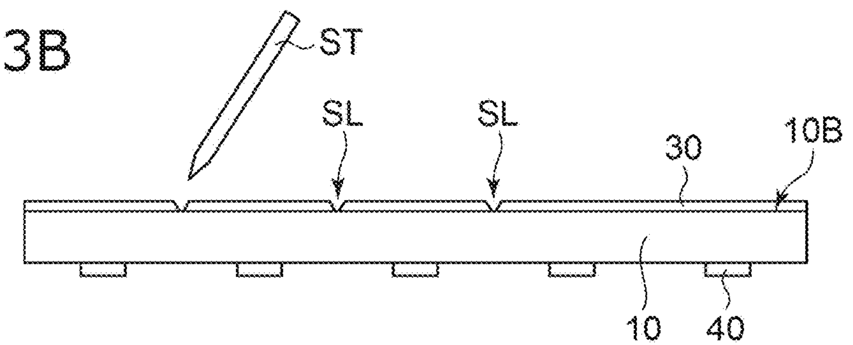



FIG. 3C

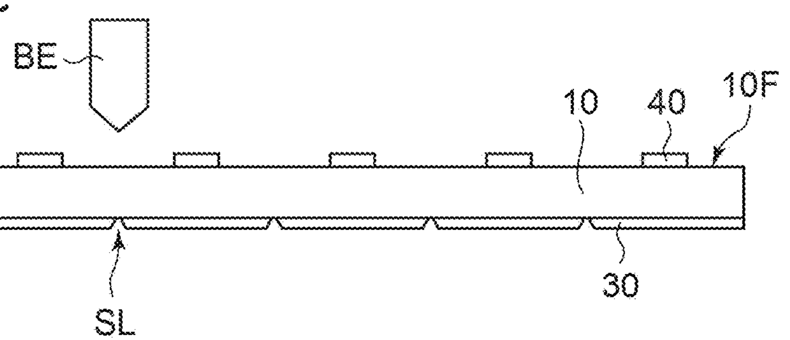



FIG. 3D

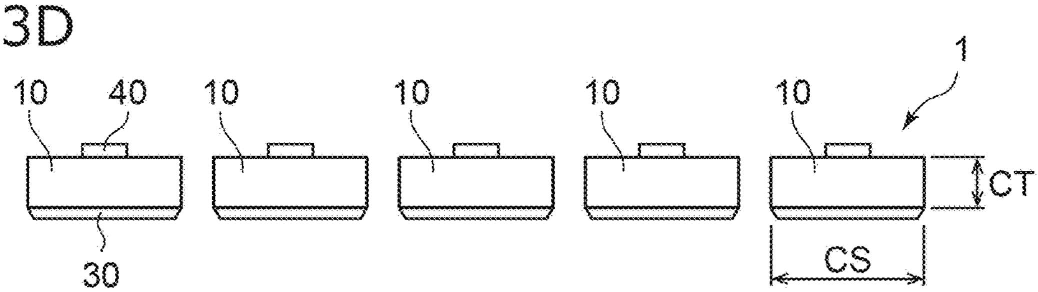


FIG. 4A

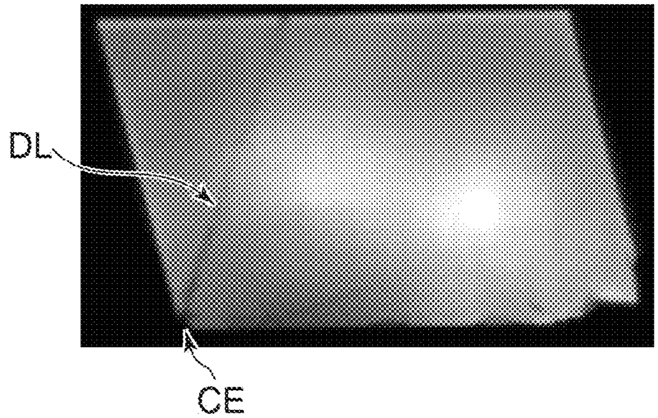



FIG. 4B

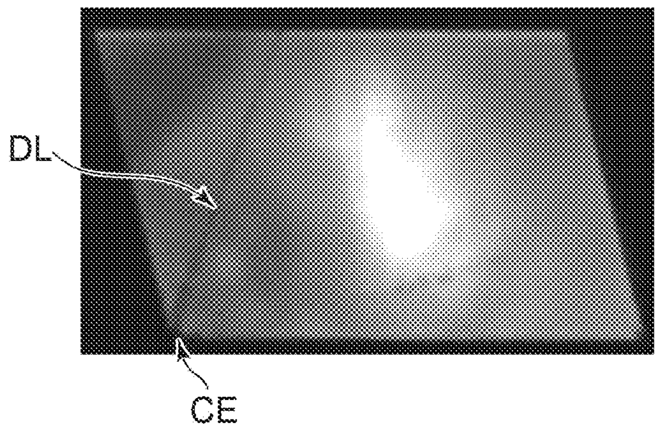



FIG. 4C

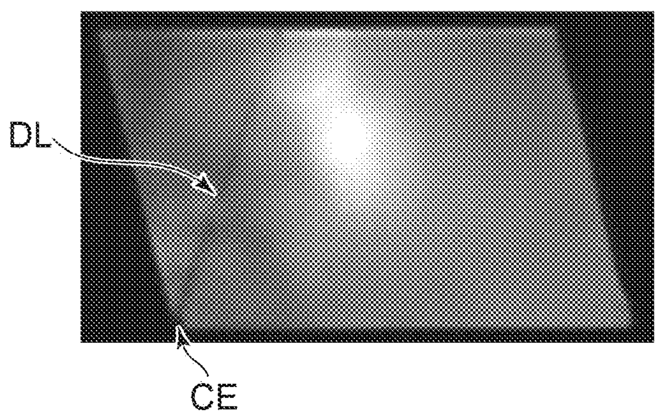
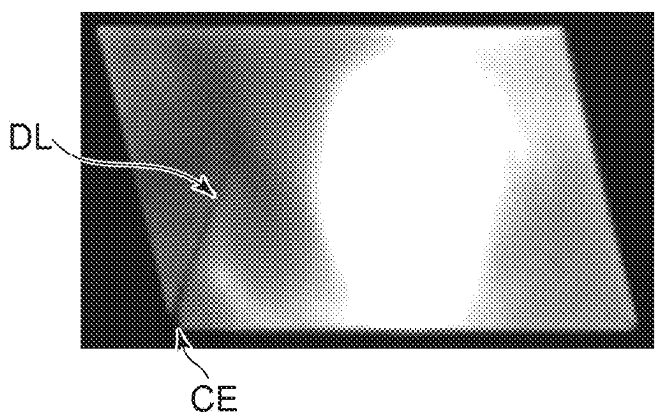



FIG. 4D



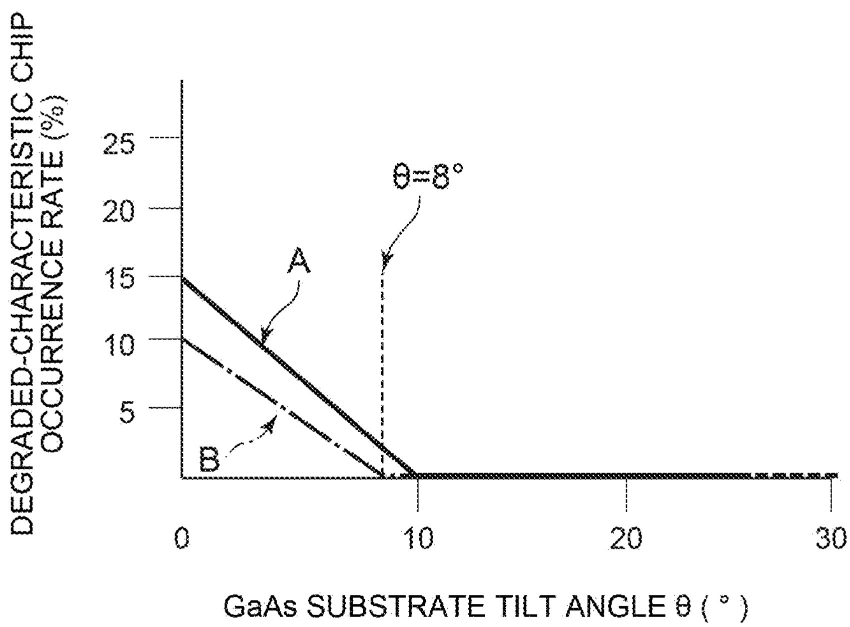

DEGRADED-CHARACTERISTIC CHIP OCCURRENCE RATE (%)
25
20
15
10
5
θ=8°
A
B
0
10
20
30
GaAs SUBSTRATE TILT ANGLE θ ( ° )

FIG. 5

FIG. 6A

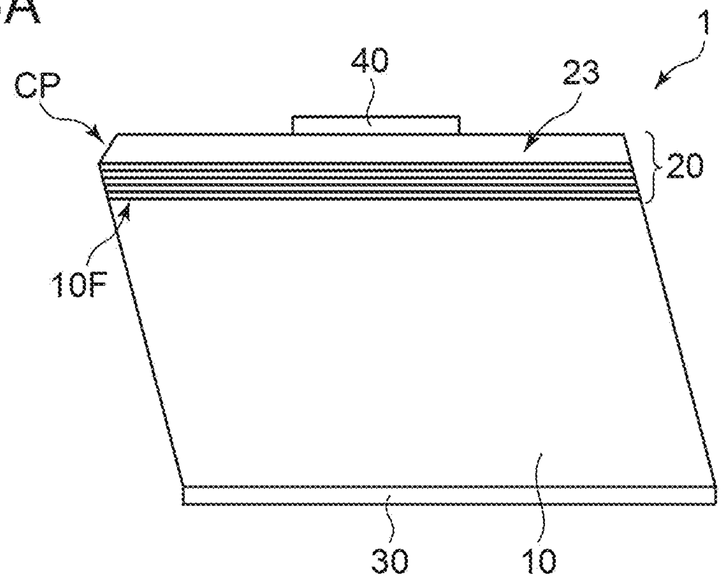



FIG. 6B

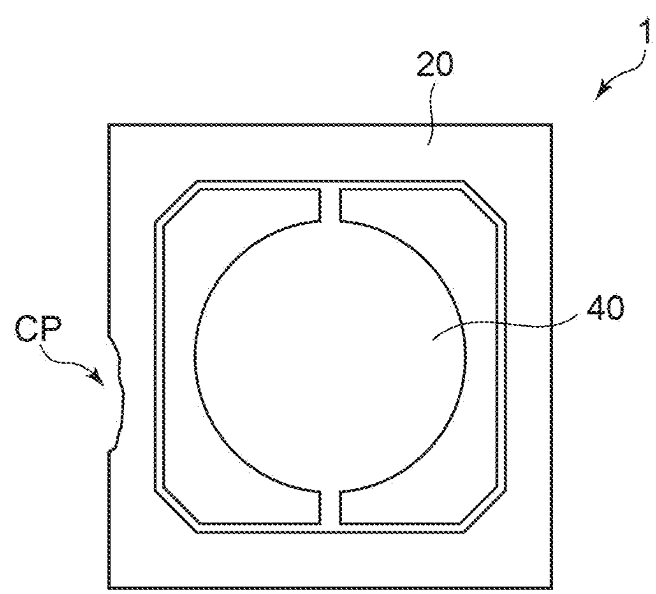


FIG. 7

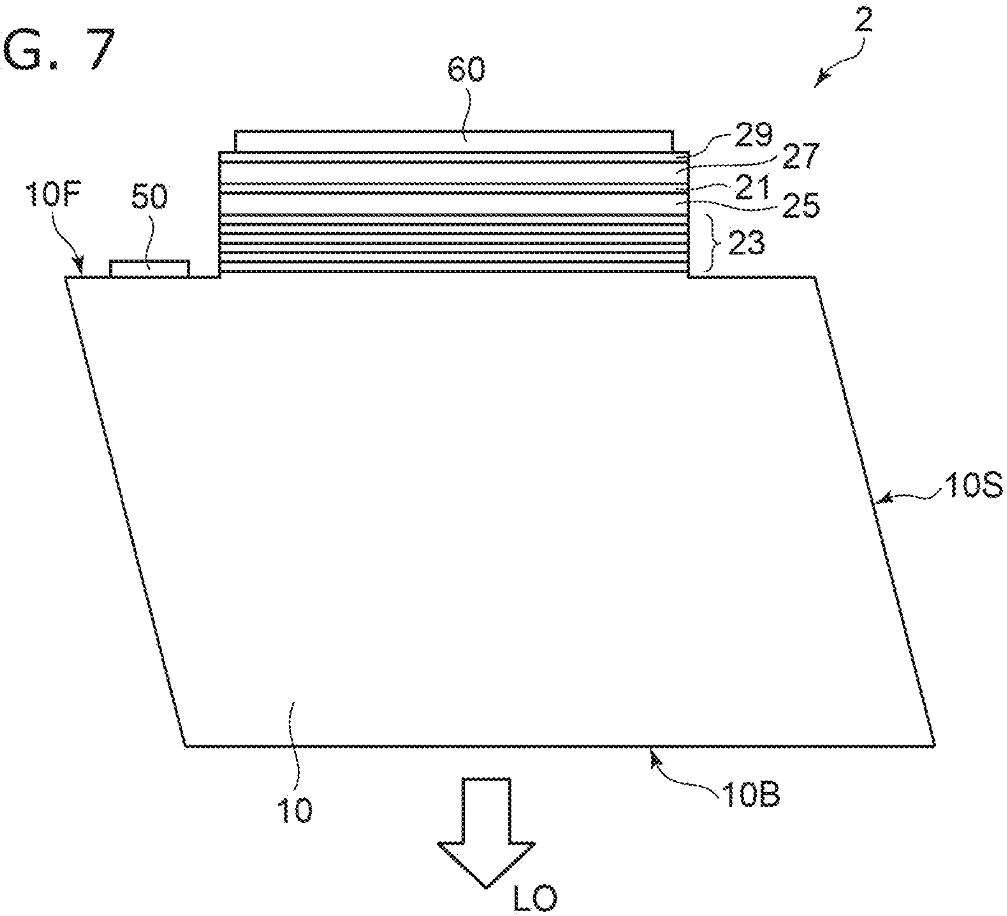
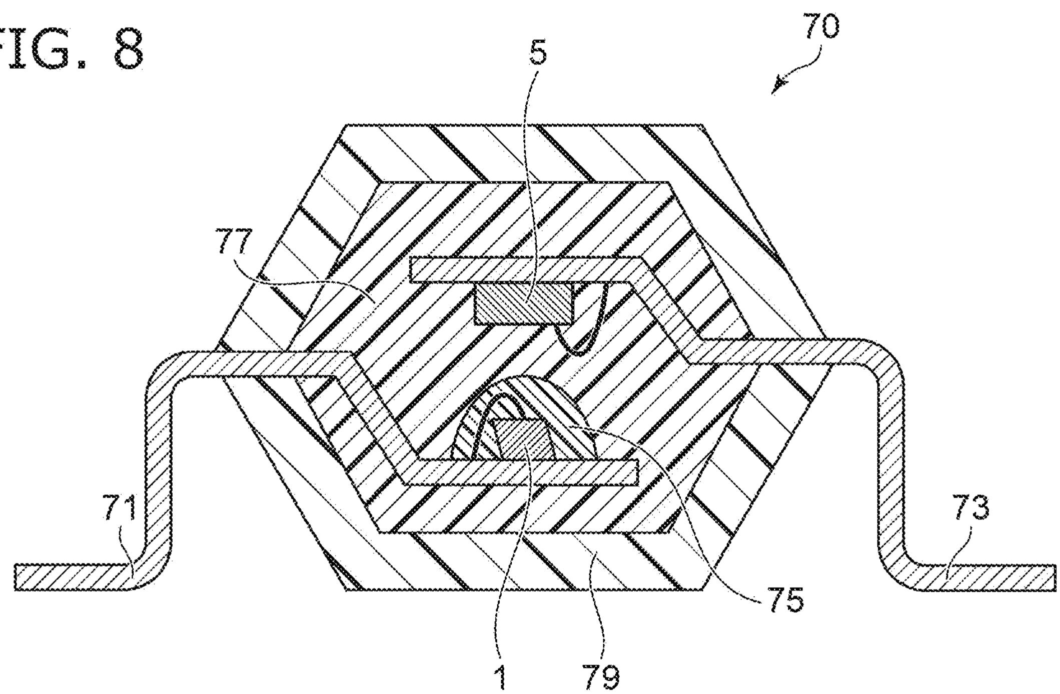



FIG. 8



# SEMICONDUCTOR LIGHT-EMITTING DEVICE AND PHOTOCOUPLER

## CROSS-REFERENCE TO RELATED APPLICATIONS

This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-008908, filed on Jan. 24, 2022; the entire contents of which are incorporated herein by reference.

## FIELD

Embodiments relate to a semiconductor light-emitting device and a photocoupler.

## BACKGROUND

A semiconductor light-emitting device is required to have high reliability. Stable operation and long life under high-temperature operating environments such as 105° C. is important for automotive applications and the like.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. **1**A and **1**B are schematic cross-sectional views showing a semiconductor light-emitting device according to an embodiment;

FIG. **2** is a schematic cross-sectional view showing the light-emitting layer of the semiconductor light-emitting device according to the embodiment;

FIGS. **3**A to **3**D are schematic cross-sectional views showing manufacturing processes of the semiconductor light-emitting device according to the embodiment;

FIGS. **4**A to **4**D are cathodoluminescence (CL) images of the semiconductor light-emitting device according to the embodiment;

FIG. **5** is a graph showing a characteristic of the semiconductor light-emitting device according to the embodiment;

FIGS. **6**A and **6**B are schematic views illustrating the semiconductor light-emitting device according to the embodiment;

FIG. **7** is a schematic cross-sectional view showing a semiconductor light-emitting device according to a modification of the embodiment; and

FIG. **8** is a schematic cross-sectional view showing a photocoupler using the semiconductor light-emitting device according to the embodiment.

## DETAILED DESCRIPTION

According to one embodiment, a semiconductor light-emitting device includes a GaAs (gallium arsenide) substrate of a cubic crystal, a light-emitting layer and a multi-semiconductor layer. The light-emitting layer being provided on the GaAs substrate. The light-emitting layer includes InGaAs (indium gallium arsenide) represented by a compositional formula InxGa1-xAs (0<x<1). The multi-semiconductor layer being provided on a front surface of the GaAs substrate between the GaAs substrate and the light-emitting layer. The multi-semiconductor layer is tilted with respect to a (100) plane of the cubic crystal. The multi-semiconductor layer includes a first layer and a second layer. The first and second layers are alternately stacked in a direction perpendicular to the front surface of the GaAs substrate. The first layer is different in composition from the second layer.



Embodiments will now be described with reference to the drawings. The same portions inside the drawings are marked with the same numerals; a detailed description is omitted as appropriate; and the different portions are described. The drawings are schematic or conceptual; and the relationships between the thicknesses and widths of portions, the proportions of sizes between portions, etc., are not necessarily the same as the actual values thereof. The dimensions and/or the proportions may be illustrated differently between the drawings, even in the case where the same portion is illustrated.

There are cases where the dispositions of the components are described using the directions of XYZ axes shown in the drawings. The X-axis, the Y-axis, and the Z-axis are orthogonal to each other. Hereinbelow, the directions of the X-axis, the Y-axis, and the Z-axis are described as an X-direction, a Y-direction, and a Z-direction. Also, there are cases where the Z-direction is described as upward and the direction opposite to the Z-direction is described as downward.

FIGS. **1**A and **1**B are schematic cross-sectional views showing a semiconductor light-emitting device **1** according to an embodiment. FIG. **1**A is a cross-sectional view of the chip. FIG. **1**B is a partial cross-sectional view showing the region surrounded with the broken line in FIG. **1**A.

The semiconductor light-emitting device **1** is, for example, a light-emitting diode that emits near-infrared light. The semiconductor light-emitting device **1** includes a substrate **10** of cubic-crystal gallium arsenide (GaAs), an epitaxial layer **20**, a first electrode **30**, and a second electrode **40**.

The GaAs substrate **10** has, for example, n-type conductivity. The GaAs substrate **10** includes a front surface **10**F. The front surface **10**F is tilted, for example, with respect to the (100) plane of the cubic crystal toward the (011) plane. The epitaxial layer **20** is provided on the front surface **10**F of the GaAs substrate **10**.

As shown in FIG. **1**A, the GaAs substrate **10** has a side surface **10**S that connects the front surface **10**F and a back surface **10**B. The side surface **10**S is, for example, the (011) plane. The semiconductor light-emitting device **1** has the cross-sectional shape of, for example, a parallelogram in a cross section parallel to the (0-11) plane. When the front surface **10**F has the tilt angle θ toward the (011) plane, the (1-1-1) plane of the cubic crystal is tilted θ+about 55° with respect to the back surface **10**B.

The first electrode **30** is provided on the back surface **10**B of the GaAs substrate **10**. The first electrode **30** is, for example, an n-side electrode. The back surface **10**B of the GaAs substrate **10** is positioned at the side opposite to the front surface **10**F and is tilted with respect to the (100) plane.

The second electrode **40** is provided on the epitaxial layer **20**. The second electrode **40** is, for example, a p-side electrode. The second electrode **40** is partially provided on the epitaxial layer **20**.

As shown in FIG. **1**B, the epitaxial layer **20** includes a light-emitting layer **21**, a multi-semiconductor layer **23**, an n-type cladding layer **25**, a p-type cladding layer **27**, and a p-type contact layer **29**.

The light-emitting layer **21** includes indium gallium arsenide (hereinbelow, InGaAs) represented by the compositional formula $In_xGa_{1-x}As$ (0<x<1). InGaAs has lower rigidity (hardness) and a smaller Young's modulus than the GaAs substrate **10**. InGaAs has a larger lattice constant than a lattice constant of the GaAs substrate **10** and provides, for example, a strained quantum well structure. The multi-semiconductor layer **23** is provided between the GaAs substrate **10** and the light-emitting layer **21**.

The multi-semiconductor layer **23** is provided on the front surface **10**F of the GaAs substrate **10**. The multi-semiconductor layer **23** has, for example, n-type conductivity. The multi-semiconductor layer **23** includes at least one first layer **23***a* and at least one second layer **23***b* alternately stacked in a direction perpendicular to the front surface **10**F of the GaAs substrate **10**. The first layer **23***a* is different in a composition from the second layer **23***b*. The first layer **23***a* is different in a rigidity (hardness) from the second layer **23***b*.

The n-type cladding layer **25** is provided between the light-emitting layer **21** and the multi-semiconductor layer **23**. The n-type cladding layer **25** includes, for example, aluminum gallium arsenide (hereinbelow, AlGaAs) represented by the compositional formula $Al_zGa_{1-z}As$ ($0<z<1$).

The p-type cladding layer **27** is provided on the light-emitting layer **21**. The p-type cladding layer **27** includes, for example, AlGaAs. The light-emitting layer **21** is provided between the n-type cladding layer **25** and the p-type cladding layer **27**.

The p-type contact layer **29** is provided on the p-type cladding layer **27**. The p-type contact layer **29** includes, for example, GaAs. The second electrode **40** is provided on the p-type contact layer **29**. The second electrode **40** is connected to the p-type contact layer **29** with, for example, an ohmic connection.

FIG. **2** is a schematic cross-sectional view showing the light-emitting layer **21** of the semiconductor light-emitting device **1** according to the embodiment. The light-emitting layer **21** includes a quantum well layer **21***w* and a barrier layer **21***b*. The quantum well layer **21***w* and the barrier layer **21***b* are alternately stacked in the direction from the n-type cladding layer **25** toward the p-type cladding layer **27**, i.e., in a direction perpendicular to the front surface **10**F of the GaAs substrate **10** (see FIG. **1**B). The quantum well layer **21***w* is provided between two adjacent barrier layers **21***b*. The quantum well layer **21***w* includes InGaAs. The barrier layers **21***b* include, for example, AlGaAs. The quantum well structure of InGaAs/AlGaAs provides a strained quantum well structure due to the lattice mismatch between the quantum well layer **21***w* and the GaAs substrate **10**. The light emission wavelength thereof is, for example, 950 nm.

FIGS. **3**A to **3**D are schematic cross-sectional views showing manufacturing processes of the semiconductor light-emitting device **1** according to the embodiment. FIGS. **3**A to **3**D illustrate the chip singulation process of the semiconductor light-emitting device **1**.

As shown in FIG. **3**A, the first electrode **30** is formed on the back surface **10**B of the GaAs substrate **10**. The first electrode **30** is provided, for example, on the entire surface of the back surface **10**B. The first electrode **30** is connected to the GaAs substrate **10** with an ohmic connection. The first electrode includes, for example, gold (Au) or silver (Ag) and serves as a reflective film of the light radiated from the light-emitting layer **21**.

The multiple second electrodes **40** are formed at the front surface **10**F side of the GaAs substrate **10**. The second electrodes **40** are provided on the epitaxial layer **20** (see FIG. **1**A) and are electrically connected to the p-type contact layer **29**.

As shown in FIG. **3**B, a scribe line SL is formed using a scriber ST in the back surface **10**B side of the GaAs substrate **10**. The scriber ST includes a diamond blade at the tip thereof. The scriber ST can form the groove-shaped scribe line SL to subdivide the first electrode **30**. The groove-shaped scribe line SL reaches the GaAs substrate **10** at the back surface **10**B side thereof. The scribe line SL is provided between two adjacent second electrodes **40** when viewed in a direction perpendicular to the back surface **10**B.

As shown in FIG. **3**C, the GaAs substrate **10** is cleaved using a breaking blade BE at the front surface **10**F side of the GaAs substrate **10**. The breaking blade BE is pressed onto the front surface **10**F at the position opposite to the scribe line SL. The GaAs substrate **10** is cleaved by the pressing of the breaking blade BE, and the scribe line SL serves as a starting point.

As shown in FIG. **3**D, the GaAs substrate **10** is subdivided into multiple chips (semiconductor light-emitting devices **1**). the semiconductor light-emitting device **1** has a chip size CS of, for example, 200 micrometers (μm). A chip thickness CT is, for example, 150 μm.

In such a chip dicing method, a mechanical damage such as micro defects and/or fracture layers is formed at the bottom of the scribe line SL that is formed in the back surface **10**B of the GaAs substrate **10** and serves as the starting point of the cleaving. Therefore, the mechanical damage remains in the chip after the dicing. Although it is possible to remove such mechanical damage by, for example, etching the GaAs substrate **10** after the chip dicing, it may be difficult to remove all of the micro defects.

FIGS. **4**A to **4**D are cathodoluminescence (CL) images of the semiconductor light-emitting device **1** according to the embodiment. The chip used in the CL measurement was selected after the accelerated aging test for 500 hours performed under high temperature and high current conditions such as atmosphere temperature of 125° C. and operating current of 120 mA. The selected chip has an operating voltage for a prescribed light output which is changed more than 10% after the reliability test.

FIGS. **4**A to **4**D show CL images of the chip side surface (see FIG. **1**A). The CL measurement observes a luminous phenomenon due to recombination of carriers (electrons and holes) excited by electron beam. Non-radiative recombination also occurs in which the excited carriers are trapped in defects. That is, when the crystal includes a defect, light is not emitted therearound.

As seen in FIG. **4**A, a dark line DL extends to the front surface of the epitaxial layer **20** from the corner at which the back surface **10**B of the GaAs substrate **10** is connected to the chip cleavage plane, i.e., the (011) plane (see FIG. **1**A). The dark line DL starts to extend from the corner at which the scribe line SL remains. The dark line extends along, for example, the (1-1-1) plane.

FIG. **4**B is a CL image observed after etching the chip side surface about 30 μm. Also, in such a case, the dark line DL appears and extends to the front surface of the epitaxial layer **20** from the starting point of the corner at which the back surface **10**B of the GaAs substrate **10** is connected to the (011) plane.

FIG. **4**C is a CL image observed after further etching the chip side surface about 30 μm (i.e., for a total etching of 60 μm). The dark line DL extends to the front surface of the epitaxial layer **20** from the corner at which the back surface **10**B of the GaAs substrate **10** is connected to the (011) plane.

FIG. **4**D is a CL image observed after further etching the chip side surface about 30 μm (for a total etching of 90 μm). The dark line DL extends into the GaAs substrate **10** from the starting point of the corner at which the back surface **10**B of the GaAs substrate **10** is connected to the (011) plane.

These CL images show that the crystal defect spreads in a plane inside the GaAs substrate **10** from the corner at which the back surface **10**B is connect to the cleavage plane of the GaAs substrate **10**. In other words, the crystal defect is a planar dislocation along the (1-1-1) plane. In the CL image of FIG. **4**D, the dark line DL does not reach the epitaxial layer **20**; and the dislocation appears to extend from the starting point of the corner at which the back surface **10**B is connected to the cleavage plane. In other words, the dislocation is found to extend inside the GaAs substrate **10** from the scribe line SL (see FIG. **3**C) that is serves as the starting point. Such a dislocation gradually extends due to, for example, temperature changes in the operating environment. When the chip is sealed with a resin, such an extension of the dislocation may be accelerated by the resin stress.

Thus, in the accelerated aging test under high temperature and high current, there may be a case where a crystal defect remaining at the scribe line SL spreads over time and reaches the epitaxial layer **20**. The light emission characteristics are degraded thereby. The test conditions described above assume a harsh environment of use, e.g., automotive applications. In such an environment, dislocation propagation that would not occur in a normal application is observed.

Generally, a reliability test is conducted under, for example, an operating current of 20 mA and a conduction time of 10000 hours at room temperature. Under such conditions, luminance degradation of, for example, 10% or more is not found. In contrast, when the reliability test is performed at the high temperature after sealing the chip with an epoxy resin, there may be a case where luminance degradation occurs due to the resin stress. It is possible to suppress such luminance degradation due to the resin stress by using a double resin-sealed structure (see FIG. **8**) in which the chip is sealed with a silicone resin that is an inelastic resin having low plastic deformation, and then, sealed with an epoxy resin. However, even in the double resin-sealed structure, there may be a case where luminance degradation occurs in the accelerated aging test under high temperature and high current conditions as described above.

The multi-semiconductor layer **23** in the epitaxial layer **20** suppresses such dislocation propagation toward the light-emitting layer **21** by combining the first layer **23***a* and the second layer **23***b* that have different rigidities; and it is possible to improve the reliability of the semiconductor light-emitting device **1**. The multi-semiconductor layer **23** changes, for example, a propagation direction of dislocation at each interface between the first layers **23***a* and the second layers **23***b* so that the dislocation does not reach the light-emitting layer **21**. Or, the multi-semiconductor layer **23** can annihilate dislocations by combining each other. Under harsh accelerated conditions, however, such a dislocation suppression effect may be insufficient.

Example 1

The multi-semiconductor layer **23** (see FIG. **1**B) can include, for example, indium aluminum phosphide (hereinbelow, InAlP) represented by the compositional formula $In_zAl_{1-z}P$ ($0<z<1$) and either GaAs or AlGaAs represented by the compositional formula $Al_vGa_{1-v}As$ ($0\leq v<1$). The first layer **23***a* of the multi-semiconductor layer **23** is taken to be, for example, InAlP; and the second layer **23***b* is taken to be, for example, GaAs ($v=0$). The first and second layers **23***a* and **23***b* each have a layer thickness of 50 nanometers (nm). The multi-semiconductor layer **23** includes, for example, ten pairs of the first layer **23***a* and the second layer **23***b*. A similar advantage may be obtained even when the second layer **23***b* is AlGaAs ($0<v<1$).

The multi-semiconductor layer **23** is configured to transmit the light radiated from the light-emitting layer **21**. The radiated light passes through the multi-semiconductor layer **23** and propagates inside the GaAs substrate **10**. Thereby, the entire chip can be luminous. As seen in conventional art, in which a light-emitting device is provided with a Bragg reflector, the light is emitted with high directivity in the direction from the multi-semiconductor layer **23** toward the second electrode **40**. In such a case, the multi-semiconductor layer **23** is configured to reflect the light radiated by the light-emitting layer **21**. In an application such as a photocoupler or the like, however, a light-emitting device preferably has low directivity. That is, it is preferable for the multi-semiconductor layer **23** to transmit the light radiated by the light-emitting layer **21**. In other words, the light of the light-emitting layer **21** is preferably radiated not only from the upper surface of the chip but also from the side surfaces so that the entire chip is luminous.

Example 2

The multi-semiconductor layer **23** (see FIG. **1**B) can include, for example, InGaAs and gallium arsenide phosphide (hereinbelow, GaAsP) represented by the compositional formula $GaAs_wP_{1-w}$ ($0<w<1$). The first layer **23***a* is taken to be $In_{0.1}Ga_{0.9}As$; and the second layer **23***b* is taken to be $GaAs_{0.9}P_{0.1}$. The first and second layers **23***a* and **23***b* each have a layer thickness of 10 nm. The multi-semiconductor layer **23** includes ten pairs of the first layer **23***a* and the second layer **23***b*.

The lattice constant of InGaAs has a magnitude relationship with respect to the lattice constant of GaAs that is the opposite magnitude relationship of the lattice constant GaAsP with respect to the lattice constant of GaAs. Also, InGaAs and GaAsP have different linear expansion coefficients. Therefore, it is possible in the multi-semiconductor layer **23** to compensate the lattice constant difference by the lattice strain within the elastic limit; and crystal defects due to the lattice mismatch do not occur. In the example, the elastic strain inside the multi-semiconductor layer **23** provides a large advantage of changing the propagation direction of the dislocation at the InGaAs/GaAsP interface. Thus, compared to the multi-semiconductor layer **23** of the example 1, the dislocation propagation is effectively suppressed in the accelerated aging test. Thereby, the characteristic degradation is suppressed, and the failure chips can be reduced.

FIG. **5** is a graph showing a characteristic of the semiconductor light-emitting device **1** according to the embodiment. The horizontal axis is the tilt angle θ of the GaAs substrate **10** (see FIG. **1**A). The vertical axis is the occurrence rate of degraded-characteristic chips in the accelerated aging test. In the figure, the graph shown by "A" illustrates a characteristic when the multi-semiconductor layer **23** of the example 1 is provided. The graph shown by "B" illustrates a characteristic when the multi-semiconductor layer **23** of the example 2 is provided.

In the example 1, the multi-semiconductor layer **23** includes InAlP and AlGaAs, and the degraded-characteristic chip occurrence rate is about 15% when the tilt angle θ is zero. The degraded-characteristic chip occurrence rate is 0% when the tilt angle θ is not less than 10°.

The same results are obtained even when the Al composition v of $Al_vGa_{1-v}As$ is changed from 0 to 0.2. Thus, it is found that there is tolerance of Al composition when AlGaAs is combined with InAlP, because of the rigidity of AlGaAs not greatly changed depending on the Al composition v.

In the example 2, the multi-semiconductor layer **23** includes InGaAs and GaAsP, the degraded-characteristic chip occurrence rate is about 10% when the tilt angle θ is zero, and the degraded-characteristic chip occurrence rate is 0% when the tilt angle θ is not less than 8°.

The propagation plane (1-1-1) of the dislocation has the tilt angle of θ+about 55° with respect to the back surface **10**B of the GaAs substrate **10** (see FIG. **1**A); and the penetration angle of the dislocation toward the multi-semiconductor layer **23** increases as the tilt angle θ increases. In other words, the larger penetration angle toward the multi-semiconductor layer **23** increases the change in the propagation direction of the dislocation, and suppresses the influence on the light-emitting layer **21**. As the tilt of the front surface **10**F in the GaAs substrate **10** is increased with respect to the (100) plane, the epitaxial layer **20** is found to have the improved crystallinity such as an improved surface morphology and like. Also, the combination of such improvements may affect the suppression of the dislocation propagation.

Thus, the reliability of the semiconductor light-emitting device **1** can be improved by tilting the front surface **10**F of the GaAs substrate **10** with respect to the (100) plane. Moreover, the reliability can be ensured in harsher environments by providing the GaAs substrate **10** with the tilt angle θ of not less than 8°, and more preferably not less than 10°. On the other hand, it is difficult to obtain an epitaxial layer with few dislocations when the tilt angle θ reaches or exceeds 25°. Accordingly, the tilt angle θ is preferably not less than 8° and not more than 25°, and more preferably not less than 10° and not more than 25°.

FIGS. **6**A and **6**B are schematic views illustrating the semiconductor light-emitting device **1** according to the embodiment. FIG. **6**A is a cross-sectional view of the chip; and FIG. **6**B is a plan view showing the chip surface.

As shown in FIG. **6**A, the semiconductor light-emitting device **1** may have a crack CP in the edge of the upper surface of the epitaxial layer **20**. The crack CP easily occurs due to the acute interior angle of the corner where the upper surface of the epitaxial layer **20** and the side surface **10**S, i.e., the cleavage plane, cross. The acute interior angle is due to the GaAs substrate **10** being tilted with respect to the (100) plane. Also, the multi-semiconductor layer **23** is one origin causing such cracks CP to occur because of the different rigidities in the first and second layers **23***a* and **23***b*.

As shown in FIG. **6**B, the crack CP is observed as a recess in one side of the chip upper surface. In the chip dicing process shown in FIGS. **3**A to **3**D, when the GaAs substrate **10** and the epitaxial layer **20** are divided along the cleavage plane, continuous cleaving from the scribe line SL to the upper surface of the epitaxial layer **20** is inhibited at the position of the multi-semiconductor layer **23**. Such inhibition induces partially chipped outer edges at the upper surface of the epitaxial layer **20**.

FIG. **7** is a schematic cross-sectional view showing a semiconductor light-emitting device **2** according to a modification of the embodiment. The semiconductor light-emitting device **2** includes the GaAs substrate **10**, the light-emitting layer **21**, the multi-semiconductor layer **23**, a first electrode **50**, and a second electrode **60**.

As shown in FIG. **7**, the epitaxial layer **20** includes the light-emitting layer **21**, the multi-semiconductor layer **23**, the n-type cladding layer **25**, the p-type cladding layer **27** and the p-type contact layer **29**, and is provided with a mesa structure. The first electrode **50** is provided on the front surface **10**F of the GaAs substrate **10** that is exposed by mesa-patterning that reaches the GaAs substrate **10**. The first electrode **50** is an n-side electrode. Alternatively, the mesa-patterning may be performed to reach the n-type cladding layer **25**, and the first electrode **50** is provided on the n-type cladding layer **25** that is exposed by the mesa-structure.

The second electrode **60** is provided on the p-type contact layer **29**. The second electrode **60** is a p-side electrode. The second electrode **60** includes, for example, gold (Au) or silver (Ag) and is configured to reflect the light radiated from the light-emitting layer **21** toward the second electrode **60** so that the light is emitted from the back surface **10**B of the GaAs substrate **10**.

In the semiconductor light-emitting device **2**, the output light LO that is radiated in vertical directions from the light-emitting layer **21** propagates through the multi-semiconductor layer **23** and through the GaAs substrate **10**. the output light LO is externally emitted from the back surface **10**B of the GaAs substrate **10**. Also, in the example, the multi-semiconductor layer **23** is configured to transmit the output light LO radiated from the light-emitting layer **21**.

FIG. **8** is a schematic cross-sectional view showing a photocoupler **70** that comprises the semiconductor light-emitting device **1** according to the embodiment. The photocoupler **70** includes the semiconductor light-emitting device **1** and a light-receiving device **5**. The semiconductor light-emitting device **1** and the light-receiving device **5** face each other and are optically coupled. In other words, the light-receiving device **5** detects the light emitted from the semiconductor light-emitting device **1**.

As shown in FIG. **8**, the semiconductor light-emitting device **1** is mounted on an input-side lead **71** and electrically connected thereto. The light-receiving device **5** is mounted on an output-side lead **73** and electrically connected thereto. The photocoupler **70** includes multiple input-side leads **71** and multiple output-side leads **73**. The photocoupler **70** includes, for example, the input-side lead **71** that is electrically connected to the first electrode **30** of the semiconductor light-emitting device (see FIG. **1**A) and another input-side lead **71** that is electrically connected to the second electrode **40** (see FIG. **1**A) via a metal wire. The multiple output-side leads **73** include a pair of output-side leads **73** connected to an anode and a cathode of the light-receiving device, respectively.

The semiconductor light-emitting device **1** is sealed inside a first resin **75**. The first resin **75** is, for example, a silicone resin. The first resin **75** transmits light emitted from the semiconductor light-emitting device **1**.

The input-side lead **71** and the output-side lead **73** are arranged such that the semiconductor light-emitting device **1** faces the light-receiving device **5**. Then, a second resin **77** is molded to cover each portion of the input-side lead **71** and the output-side lead **73**. The semiconductor light-emitting device **1** is mounted on the portion of the input-side lead **71**, and the light-receiving device **5** is mounted on the portion of the output-side lead **73**. The second resin **77** covers the semiconductor light-emitting device **1** and the light-receiving device **5**. The second resin **77** covers the light emitting device **1** with the first resin **75** interposed. The second resin **77** transmits the light emitted from the semiconductor light-emitting device **1**. The second resin **77** is, for example, an epoxy resin.

A third resin **79** is molded to cover the second resin **77**. The third resin **79** shields the light emitted from the semiconductor light-emitting device **1**. The third resin **79** is, for example, an epoxy resin that includes carbon.

In the photocoupler **70**, the semiconductor light-emitting device **1** is sealed with the first resin **75** that is an inelastic resin. The semiconductor light-emitting device **1** also is sealed via the first resin **75** with the second resin **77**. The second resin **77** has a higher hardness than the first resin **75**. By using such a double sealing structure, it is possible to reduce the resin stress applied to the semiconductor light-emitting device **1**, and improve the reliability thereof.

As described above, in the semiconductor light-emitting devices **1** and **2**, the multi-semiconductor layer **23** is provided between the GsAs substrate **10** and the light-emitting layer **21**, and the front surface **10**F of the GaAs substrate **10** is tilted with respect to the (100) plane. Thereby, it is possible in the semiconductor light-emitting devices **1** and **2** to prevent crystal defects generated in the chip dicing process from influencing on the reliability thereof. Such an advantage is more pronounced when the light-emitting layer **21** has, for example, a lower rigidity (hardness) than the GaAs substrate **10**, and the multi-semiconductor layer **23** includes multiple layers that are mutually different in rigidities or linear expansion coefficients.

In other words, the multi-semiconductor layer **23** suppresses the propagation of dislocations from the GaAs substrate **10** toward the light-emitting layer **21**. The suppressing effect of the dislocation propagation in the multi-semiconductor layer **23** can be increased by tilting the crystal growth surface of the GaAs substrate **10** with respect to the (100) plane. Such dislocation suppression effects are more pronounced because the light-emitting layer **21** of the semiconductor light-emitting device **1** includes the strained quantum well in which light emission characteristics are easily degraded by the dislocations propagating thereinto.

Embodiments are not limited to the example described above. For example, the chips can be diced using a dicing blade instead of scribing. Even when the dicing blade is used, the cutting surface may include defects; and the dislocations extend along crystal planes equivalent to a (111) plane. Accordingly, combining the multi-semiconductor layer **23** and the tilt of the front surface **10**F of the GaAs substrate **10** is advantageous even when dicing is performed using the dicing blade.

The number of pairs of the first layer **23***a* and the second layer **23***b* in the multi-semiconductor layer **23** is not limited to ten; there may be a case where two or more pairs of the first layer **23***a* and the second layer **23***b* are sufficient. The propagation suppression of dislocations occurs at the interface between the first layer **23***a* and the second layer **23***b*. Accordingly, various modifications of the film thicknesses of the first and second layers **23***a* and **23***b* are possible.

Although the (011) plane is illustrated as the side surface **10**S of the GaAs substrate **10**, the side surface **10**S is not limited thereto. For example, crystal planes equivalent to the (011) plane, i.e., (01-1), (0-1-1), and (0-11), may be used. In other words, dislocations propagate along the (1-11) plane when the front surface **10**F of the GaAs substrate **10** is tilted with respect to the (01-1) plane. Dislocations propagate along the (111) plane when the tilt is toward the (0-1-1) plane; and dislocations propagate along the (11-1) plane when the tilt is toward the (0-11) plane.

While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the invention.

What is claimed is:

**1**. A semiconductor light-emitting device, comprising:

a GaAs (gallium arsenide) substrate of a cubic crystal;

a light-emitting layer provided on the GaAs substrate, the light-emitting layer including InGaAs (indium gallium arsenide) represented by a compositional formula $In_xGa_{1-x}As$ ($0<x<1$); and

a multi-semiconductor layer provided on a front surface of the GaAs substrate between the GaAs substrate and the light-emitting layer, the multi-semiconductor layer being tilted with respect to a (100) plane of the cubic crystal,

the multi-semiconductor layer including a first layer and a second layer, the first layer and the second layer being alternately stacked in a direction perpendicular to the front surface of the GaAs substrate, the first layer being different in composition from the second layer.

**2**. The device according to claim **1**, wherein

the front surface of the GaAs substrate is tilted 8 degrees to 25 degrees with respect to the (100) plane of the cubic crystal.

**3**. The device according to claim **2**, wherein

the front surface of the GaAs substrate is tilted from the (100) plane in a direction toward a (011) plane, the (011) plane being orthogonal to the (100) plane.

**4**. The device according to claim **1**, wherein

the GaAs substrate further includes a back surface and a side surface, the back surface being at a side opposite to the front surface, the side surface being connected to the front surface and the back surface, and

at least one of the side surface of the GaAs substrate or a corner connecting the side surface and the back surface of the GaAs substrate includes a crystal dislocation extending in a direction toward the multi-semiconductor layer.

**5**. The device according to claim **4**, wherein

the crystal dislocation does not reach the light-emitting layer.

**6**. The device according to claim **4**, wherein

the side surface of the GaAs substrate is a (011) plane of the cubic crystal or a crystal plane equivalent to the (011) plane.

**7**. The device according to claim **6**, wherein

the crystal dislocation of the GaAs substrate extends along a crystal plane equivalent to a (111) plane of the cubic crystal.

**8**. The device according to claim **1**, wherein

the multi-semiconductor layer transmits light radiated from the light-emitting layer.

**9**. The device according to claim **1**, wherein

the first layer of the multi-semiconductor layer is an InAlP (indium aluminum phosphide) layer represented by a compositional formula $In_yAl_{1-y}P$ ($0<y<1$), and

the second layer is a GaAs layer or a AlGaAs (aluminum gallium arsenide) layer represented by a compositional formula $Al_vGa_{1-v}As$ ($0\leq v<1$).

**10**. The device according to claim **1**, wherein

the first layer of the multi-semiconductor layer is an InGaAs layer, and

the second layer is a GaAsP (gallium arsenide phosphide) layer represented by a compositional formula $GaAs_wP_{1-w}$ ($0<w<1$).

**11**. The device according to claim **1**, wherein the light-emitting layer is included in an epitaxial layer, the epitaxial layer includes an upper surface positioned at a side opposite to the GaAs substrate, and the epitaxial layer includes a partial recess at an outer edge of the upper surface.

**12**. A photocoupler, comprising:

a light-emitting device;

a first resin sealing the light-emitting device; and

a second resin covering the first resin and the light-emitting device sealed by the first resin, the second resin having a hardness greater than a hardness of the first resin,

the light-emitting device including a GaAs substrate of a cubic crystal, a light-emitting layer and a multi-semiconductor layer,

the light-emitting layer being provided on the GaAs substrate, and including InGaAs represented by a compositional formula $In_xGa_{1-x}As$ ($0<x<1$),

the multi-semiconductor layer being provided on a front surface of the GaAs substrate between the GaAs substrate and the light-emitting layer, the multi-semiconductor layer being tilted with respect to a (100) plane of the cubic crystal,

the multi-semiconductor layer including a first layer and a second layer, the first layer and the second layer being alternately stacked in a direction perpendicular to the front surface of the GaAs substrate, the first layer being different in composition from the second layer.

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